ARJUNA JEE 2027

Lecture -10 | Mathematics

PERMUTATIONS AND COMBINATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A **Distribution of Identical Objects**

B **Circular Permutation**

C **Question Practice**

D



H.W. Discussion

Question [JEE Mains-2026]

[Ans. : 126]

Two players A and B play a series of games of badminton, The player, who wins 5 games first, wins the series. Assuming that no game ends in a draw, the number of ways, in which player A wins the series is _____.

(A), B

5 Matches — — — — — → $^5C_5 = 1$

6 Matches — — — — — (W) → $^5C_4 = 5$

A wins

7 Matches — — — — — — (—) → $^6C_4 = 15$

8 Matches — — — — — — — (A) → $^7C_4 = 35$

wins

9 Matches — — — — — — — — (A) → $^8C_4 = 70$

10 Matches ✗

$\text{Inf} = (1 + 5 + 15 + 35 + 70)$

$= 126.$

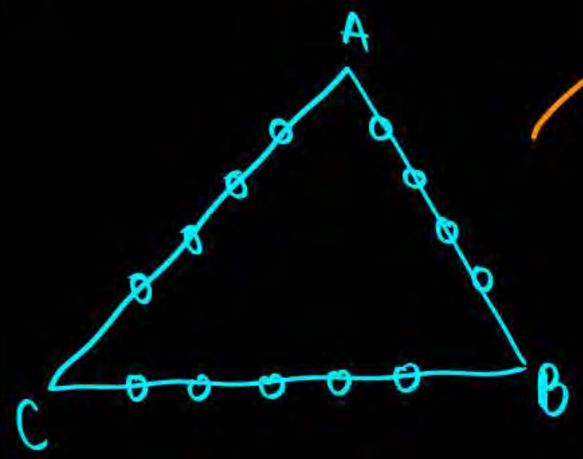
OR

$^9C_5 = 126$

Question [JEE Mains-2026]

[Ans. : 660]

Let ABC be a triangle. Consider four points p_1, p_2, p_3, p_4 on the side AB, five points p_5, p_6, p_7, p_8, p_9 on the side BC, and four points $p_{10}, p_{11}, p_{12}, p_{13}$ on the side AC. None of these points is a vertex of the triangle ABC. Then the total number of pentagons, that can be formed by taking all the vertices from the points $p_1, p_2, \ldots, p_{13}$, is



Select any 5 points

	AB	BC	CA
C-I:	2	2	1
	$^4C_2 \times {}^5C_2 \times {}^4C_1 =$ ...		
C-II:	2	1	2
	$^4C_2 \times {}^5C_1 \times {}^4C_2 =$ ...		
C-III:	1	2	2
	$^4C_1 \times {}^5C_2 \times {}^4C_2 =$ ...		

add

Question

Find the number of arrangements of letters of the word EQUATION such that vowels occur in alphabetical order.



$$^{8}C_{5} \times 1 \times 3! = 336$$

Alph. order

(A, E, I, O, U) (Q, T, N)

Question

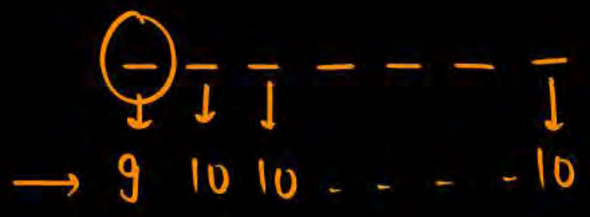
When repetition is allowed find the number of 7 digits numbers 0 to 9 (using digits)

(i) such that sum of digits is even

(ii) which are palindrome.



(9000) Ans.



Total = 9×10^6

Sum of digit is even = $\frac{9 \times 10^6}{2}$

Question [JEE Mains-2020]

The number of 4 letter words (with or without meaning) that can be formed from the eleven letters of the word EXAMINATION is _____.

Ans: 2454

A A, N N, I I, E, X, T, O, M

C-I :- all diff

X T O M

A N I E

$({}^{8}C_{4} \times 4!_{0})$ +

C-II :- 2 alike + 2 diff

A A I E

N N I O

$({}^{3}C_{1} \times {}^{7}C_{2} \times \frac{4!_{0}}{2!_{0}})$ +

C-III :- 2 alike + 2-other alike

A N N A

I I A A

$({}^{3}C_{2} \times \frac{4!_{0}}{2!_{0}2!_{0}})$

Question [JEE Mains-2026]

[Ans. : (D)]

The number of 4-letter words, with or without meaning, each consisting of two vowels and two consonants that can be formed from the letters of the word **INCONSEQUENTIAL**, without repeating any letter, is

A 2670

B 2840

C 2920

D 3600

Vowels (A, E, I, O, U)

Consonants (N, C, S, Q, T, L)

$${}^{5}C_{2} \times {}^{6}C_{2} \times (4!)$$

ex:- ENNE ✗

Question [JEE Mains-2023]

Let n be a non-negative integer. Then the number of divisors of the form $4n+1$ of the number $(10)^{10}.(11)^{11}.(13)^{13}$ is equal to _____.

81%

$N = 2^{10} \cdot 5^{10} \cdot 11^{11} \cdot 13^{13}$

$2^0 \quad 2^1 \quad 2^2 \ldots\ldots 2^{10}$

$4n+1 \quad 4n+2 \quad \underbrace{\qquad 4n \qquad}$

$\underbrace{13^0}_{4n+1} \quad \underbrace{13^1}_{4n+1} \quad \underbrace{13^2}_{4n+1} \ldots\ldots 13^{13}$

$\underbrace{5^0 \quad 5^1 \quad 5^2 \ldots\ldots 5^{10}}_{4n+1}$

$\underbrace{11^0}_{4n+1} \quad \underbrace{11^1}_{4n+3} \quad \underbrace{11^2}_{4n+1} \quad \underbrace{11^3}_{4n+3} \ldots\ldots 11^{11}$

$4n+1 \quad 4n+3 \quad 4n+1 \quad 4n+3 \ldots\ldots$

$4n+1 = (4n+1)_{2} \cdot (4n+1)_{5} \cdot (4n+1)_{11} (4n+1)_{13}$

$1 \times 11 \times 6 \times 14$

↳ 924 Ans.

LAST CLASS RECAP

1. Number of ways of distributing 5 different chocolates to 5 children such that each can get any amount is $5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 = 5^5$.

2. Number of ways of distributing 5 different chocolates to 5 children such that each get one, is $5!$.

3. Number of ways of distributing 5 different chocolates to 3 children such that each get at-least one, is _____.

$n = 5$, $r = 3$

$3^5 - {}^3C_1(2)^5 + {}^3C_2(1)^5 - {}^3C_3 0^5$

$243 - 3 \times 32 + 3 = 246 - 96 = 150$

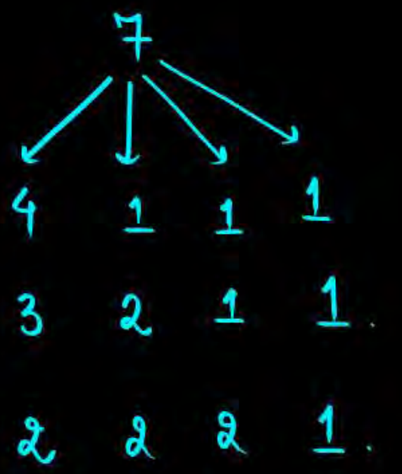


S, R, P 90 + 60

$\left(\frac{5!}{2!2!1!1!} \times 3!\right) + \left(\frac{5!}{3!1!1!} \times 3!\right)$

Example

Number of ways of distributing seven different chocolates to four children such that each gets atleast one chocolate.



$$\left(\frac{7!}{4!(1!)^3 3!} \times 4! + \frac{7!}{3!2!(1!)^2 2!} \times 4! + \frac{7!}{(2!)^3 1! 3!} \times 4!\right)$$

$\Rightarrow$ fnf = 8400.

DISTRIBUTION

Number of ways of Distributing 'n'-different things among 'r'-Persons (such that each get atleast one) =

direct formula

$$ {}^{r}C_0 (r)^n - {}^{r}C_1 \times (r-1)^n + {}^{r}C_2 (r-2)^n - {}^{r}C_3 (r-3)^n + \ldots\ldots $$



Example

Number of ways of distributing seven different chocolates to four children such that each gets atleast one chocolate.

$n = 7$
$r = 4$

$$\text{fnf} = {}^4C_0\, 4^7 - {}^4C_1\, 3^7 + {}^4C_2\, 2^7 - {}^4C_3\, 1^7 + {}^4C_4\, 0^7$$

0

$$= 8400.$$

RAPID FIRE DIBY'S

DIBY-01

Number of ways in which 8 computers of different make can be distributed among 5 schools such that each school gets at least one computer is:

(A) 63,000 (B) 126,000 (C) 151,200 (D) 189,000

[Ans. : (B)]

DIBY-02

In how many ways can a selection of 5 letters be made from the word MISSISSIPPI?

(A) 15 (B) 25 (C) 42 (D) 60

[Ans. : (B)]

DIBY-03

In how many different ways the number $N = 2^4 . 3^2 . 5^3 . 7^2 . 11^4 . 13^2$ be resolved as a product of two divisors which are relatively prime ?

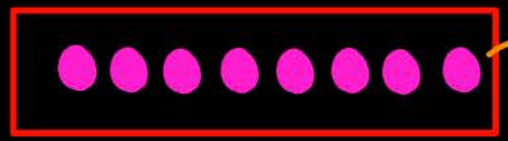
(A) 32 (B) 64 (C) 16 (D) 128

[Ans. : (A)]

Stick - Partition Method

n - id. things
↓
r - children.

Total ways $= \dfrac{(n+r-1)!}{n!(r-1)!} = {}^{n+r-1}C_{r-1}$

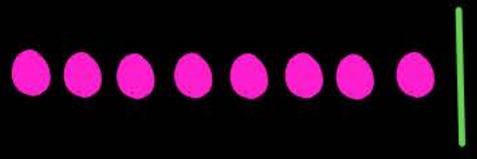


Things

4 children

'1' कम

Sticks : 3

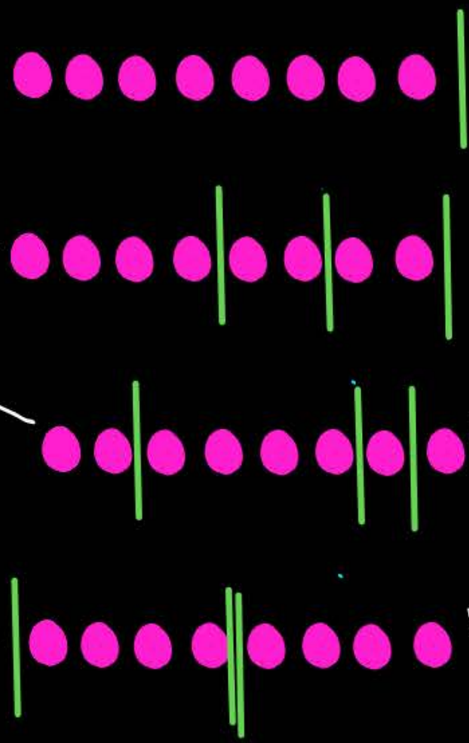


Total arrangement

$\dfrac{11!}{8!\,3!} = {}^{11}C_3$

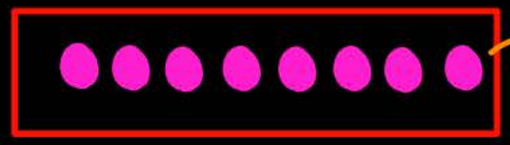
fnf.

A	B	C	D
8	0	0	0
4	2	2	0
2	4	1	1
0	4	0	4
⋮			



Stick - Partition Method

n - id. things → r - child } each gets atleast one = $^{n-1}C_{r-1}$

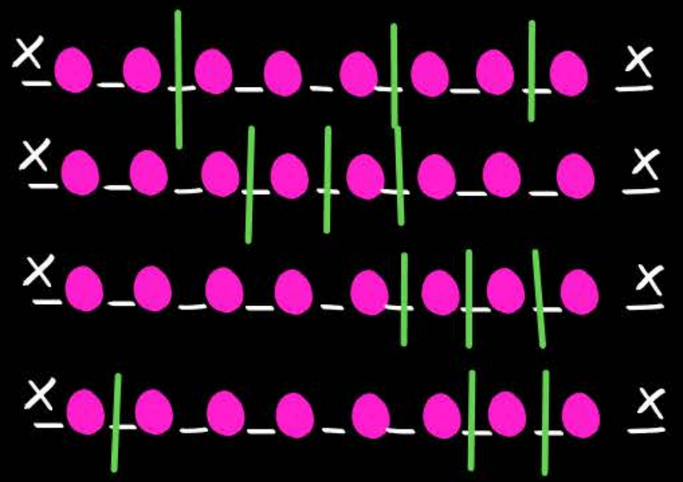


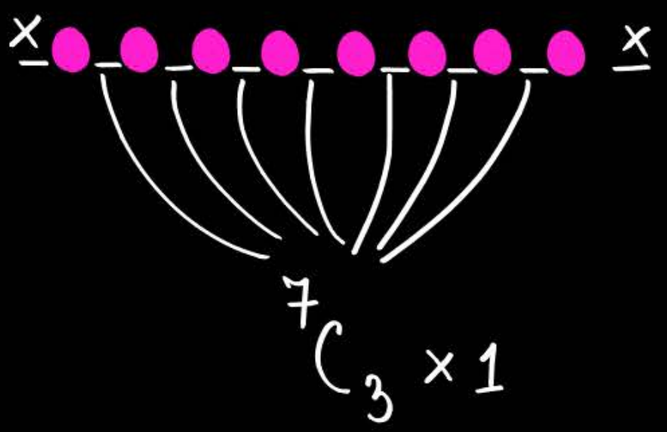
4 children (atleast one)

'1' कम

Sticks : (3)







DISTRIBUTION OF IDENTICAL OBJECTS

Question

$n = 13, \; r = 5$

Number of ways of distributing 13 identical things to 5 children such that:

(i) **No restriction (किसी को कितनी भी देदो)**

$$^{n+r-1}C_{r-1} = {}^{13+5-1}C_{5-1} = \boxed{{}^{17}C_4}$$

(ii) **each gets at least one.**

$$^{n-1}C_{r-1} = {}^{13-1}C_{5-1} = \boxed{{}^{12}C_4}$$

NUMBER OF INTEGRAL SOLUTIONS (BEGGAR's METHOD)

Equation: $x_1 + x_2 + x_3 + \cdots + x_r = n$

(r- children; n → no. of identical things)

Case-I: $(x_1, x_2, x_3, \ldots, x_r) \in N \{1, 2, 3, 4 \ldots\}$ (atleast one given to each child)

Number of integral solutions $= {}^{n-1}C_{r-1}$

Case-II: $(x_1, x_2, x_3, \ldots, x_r) \in W \{0, 1, 2, 3 \ldots\}$ (all possible dist.)

Number of integral solutions $= {}^{n+r-1}C_{r-1}$

$x + y + z + t = 18$

$n = 18$

$r = 4.$

① $x, y, z, t \in N \rightarrow {}^{n-1}C_{r-1} = {}^{17}C_3.$

② $x, y, z, t \in W. \rightarrow {}^{n+r-1}C_{r-1} = {}^{21}C_3.$

Find number of integral solutions of following equation with given conditions:

(i) $x + y + z + t = 16$, **such that** $(x \geq 1, y \geq 2, z \geq 3, t \geq 0)$

$x-1 \geq 0$, $y-2 \geq 0$, $z-3 \geq 0$, $t \geq 0$

$X \geq 0$ $Y \geq 0$ $Z \geq 0$

initial cond^{n} ✓

$x' + y' + z' + t = 10$ → all possible dist $= {}^{10+4-1}C_{4-1} = {}^{13}C_3$.

$(x-1) + (y-2) + (z-3) + t = 10$

$X + Y + Z + t = 10$, $(X, Y, Z, t \in W)$

Find number of integral solutions of following equation with given conditions:

(ii) $x + y + z + t = 16$, **such that** $(x \geq 4, y \geq -2, z \geq -3, t \geq -1)$

$x-4 \geq 0,\ y+2 \geq 0,\ z+3 \geq 0,\ t+1 \geq 0$

$(x-4) + (y+2) + (z+3) + (t+1) = 16 - 4 + 2 + 3 + 1$

$X \geq 0,\ Y \geq 0,\ Z \geq 0,\ T \geq 0.$

$X + Y + Z + T = 18$

$${}^{n+r-1}C_{r-1} = {}^{18+4-1}C_{4-1} = {}^{21}C_3 \text{ Ans.}$$

Find number of integral solutions of following equation with given conditions:

(iii) $x + y + z + t = 16$, **such that** $(x, y, z, t \in$ **odd +ve integers)**

$(2p+1) + (2q+1) + (2m+1) + (2l+1) = 16$

$p + q + m + l = 6$

all possible $= {}^{n+r-1}C_{r-1} = {}^{6+4-1}C_{4-1} = {}^{9}C_{3}$.

x	y	z	t
1	5	7	3
3	1	1	11

0	2	3	1
1	0	0	5
4	2	2	0
⋮			

Find number of integral solutions of following equation with given conditions:

(iv) $x + y + z + 4\omega = 20$, **such that** $(x, y, z, \omega \in$ **positive integers)**

(Natural No.)

C-I: $\omega = 1 \Rightarrow x + y + z = 16 \Rightarrow {}^{16-1}C_{3-1} = {}^{15}C_2$

C-II: $\omega = 2 \Rightarrow x + y + z = 12 \Rightarrow {}^{12-1}C_{3-1} = {}^{11}C_2$

C-III: $\omega = 3 \Rightarrow x + y + z = 8 \Rightarrow {}^{8-1}C_{3-1} = {}^{7}C_2$

C-IV: $\omega = 4 \Rightarrow x + y + z = 4 \Rightarrow {}^{4-1}C_{3-1} = {}^{3}C_2$

C-V: $\omega = 5 \Rightarrow x + y + z = 0 \Rightarrow$ not possible.

add = fnf.

Find number of integral solutions of following equation with given conditions:

(iv) $x + y + z + 4\omega = 20$, **such that** ($x, y, z, \omega \in$ **whole no.**)

C-I : $\omega = 1 \Rightarrow x + y + z = 16 \Rightarrow {}^{16-1+3}C_{3-1} = {}^{18}C_2$

C-II : $\omega = 2 \Rightarrow x + y + z = 12 \Rightarrow {}^{12-1+3}C_{3-1} = {}^{14}C_2$

C-III : $\omega = 3 \Rightarrow x + y + z = 8 \Rightarrow {}^{8-1+3}C_{3-1} = {}^{10}C_2$

C-IV : $\omega = 4 \Rightarrow x + y + z = 4 \Rightarrow {}^{4-1+3}C_{3-1} = {}^{6}C_2$

C-V : $\omega = 0 \Rightarrow x + y + z = 20 \Rightarrow {}^{20+3-1}C_{3-1} = {}^{22}C_2$

add = fnf.

Find number of integral solutions of following equation with given conditions:

(v) $x + y + z \leq 20$, **such that** ($x, y, z \in$ **non-negative integers**)

(Whole No.)

ex:- $10 + 0 + 10$ ✓

$5 + 6 + 0$ ✓

$1 + 1 + 1$ ✓

$0 + 0 + 0$ ✓

∴ False beggar

$x + y + z + t = 20$

10 0 7	3
5 1 4	10
0 0 0	20
9 9 2	0

all possible $= {}^{n+r-1}C_{r-1}$

$= {}^{20+4-1}C_{4-1}$

$= {}^{23}C_3$ ✓

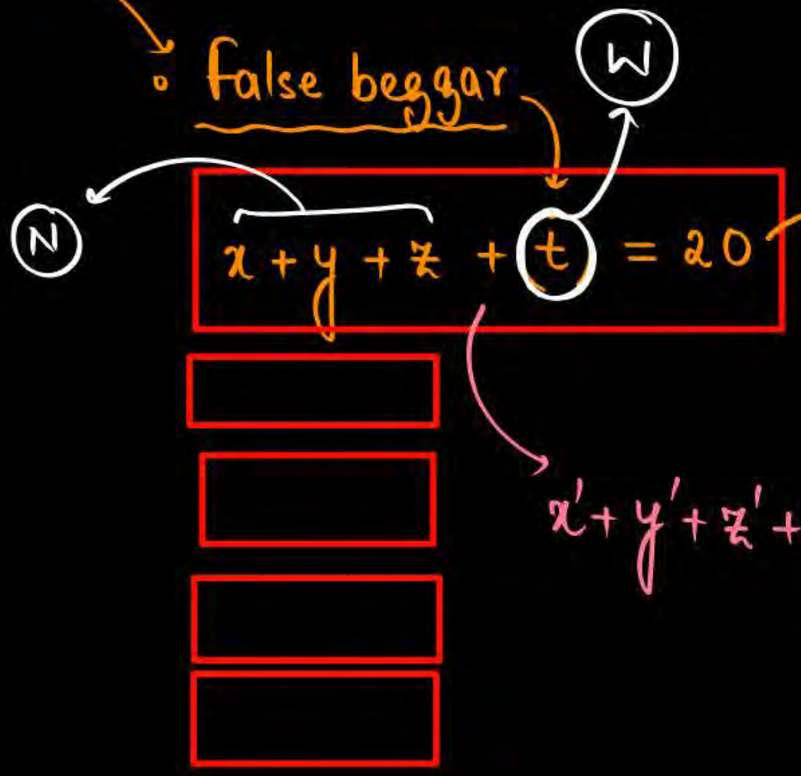
Find number of integral solutions of following equation with given conditions:

(v) $x + y + z \leq 20$, **such that** ($x, y, z \in$ Natural No.

$x+y+z+t = 20$

$x, y, z \geq 1, \; t \geq 0$.

∴ False beggar



all possible = $^{17+4-1}C_{4-1}$

$= {}^{20}C_3$.

$x' + y' + z' + t = 17$

Maha Bumper Question

Find number of integral solutions of following equation with given conditions:

(vi) $35 < x + y + z \leq 52$**, such that (**$x, y, z \in$ **positive integers)**



Acidic Question

How many words of seven letters can be formed by using the letters of the word 'SUCCESS' so that neither the two C nor any two S are together.



A 96

B 41

C 264

D 40

RAPID FIRE DIBY'S

DIBY-01

In how many ways can 30 marks be allotted to 8 questions if at least 2 marks are to be given to each question, assuming the marks are allotted in non negative integral values?

(A) ${}^{21}C_7$ (B) ${}^{22}C_7$ (C) ${}^{29}C_7$ (D) ${}^{21}C_8$

[Ans. : (A)]

DIBY-02

There are an unlimited number of Red, White, Black, and Green balls (balls are all alike except for the color). Find the number of ways in which a selection of 10 balls can be made:

(A) ${}^{13}C_4$ (B) ${}^{10}C_3$ (C) ${}^{13}C_3$ (D) ${}^{14}C_3$

[Ans. : (C)]

DIBY-03

A supermarket offers ice creams in 10 different varieties. Ice creams of each variety are identical. In how many ways can 4 ice creams be selected without any restriction?

(A) ${}^{13}C_6$ (B) ${}^{10}C_4$ (C) ${}^{13}C_5$ (D) ${}^{13}C_4$

[Ans. : (D)]

DO IT BY YOURSELF (DIBY)

DIBY-55

Find the total number of ways of selecting five letters from the work 'INDEPENDENT'.

[Ans. : 72]

DIBY-56

The number of ways five alphabets can be chosen from the alphabets of the word MATHEMATICS, where the chosen alphabets are not necessarily distinct, is equal to [JEE Mains-2024]

(A) 179 (B) 177 (C) 181 (D) 175

[Ans. : (A)]

DIBY-57

The total number of words (with or without meaning) that can be formed out of the letters of the word 'DISTRIBUTION' taken four at a time, is equal to

[JEE Mains-2024]

[Ans. : 3734]

DIBY-58

In an examination of Mathematics paper, there are 20 questions of equal marks and the question paper is divided into three sections: A, B and C. A student is required to attempt total 15 questions taking at least 4 questions from each section. If section A has 8 questions, section B has 6 questions and section C has 6 questions, then the total number of ways a student can select 15 questions is [JEE Mains-2024]

[Ans. : 11376]

DIBY-59

In how many ways 5 NEET & 5 JEE Aspirants can be seated

(a) All JEE aspirants are together

(b) All NEET aspirants are together & all JEE aspirants are together

(c) Not all JEE aspirants are together

(d) No 2 JEE aspirants are together.

[Ans. : .....]

DIBY-60

Find the number of all 6 digit numbers such that all the digits of each number are selected from the set $\{1, 2, 3, 4, 5\}$ and any digit that appears in the number appears at least twice.

[Ans. : 1405]

DIBY-61

The number of words that can be formed using all the letter of the word "CURRICULUM" in which two U's are together but separated from the third 'U' in $K(7!)$. The value of K is

(A) 7 (B) 9 (C) 12 (D) 14

[Ans. : (D)]

DIBY-62

The total number of ways in which 5 balls of different colours can be distributed among 3 persons so that each person gets at least one ball is [IIT-JEE-2012]

(A) 75 (B) 150 (C) 210 (D) 243

[Ans. : (B)]

DIBY-63

The students $S_1, S_2, \ldots, S_{10}$ are to be divided into 3 groups A, B and C such that each group has at least one students and the group C has at most 3 students. Then the total number of possibilities of forming such groups is

[JEE Mains-2021]

[Ans. : 31650]

DIBY-64

The total number of three-digit numbers, divisible by 3, which can be formed using the digits 1, 3, 5, 8 if repetition of digits is allowed, is [JEE Mains-2023]

(A) 21 (B) 22 (C) 18 (D) 20

[Ans. : (B)]

DIBY-65

Let the set $S = \{2, 4, 8, 16, \ldots. 512\}$ be partitioned into 3 sets A, B, C with equal number of elements such that $A \cup B \cup C = S$ and $A \cap B = B \cap C = A \cap C = \phi$. The maximum number of such possible partitions of S is equal to [JEE Mains-2024]

(A) 1640 (B) 1520 (C) 1710 (D) 1680

[Ans. : (D)]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians